ARGUS-NCR

Delhi NCR Intelligence Platform

Complete Technical Documentation

All Phases · All Features · All Modules

Classification: UNCLASSIFIED (Demonstration Model)

Generated: 16 June 2026

Built with Next.js 16 + TypeScript + Tailwind

Architecture

Next.js 16 + React + TypeScript + Tailwind CSS
Leaflet Maps · Custom SVG Graphs · Recharts · jspdf
Node.js Data Pipelines · Cron Automation · Real-time JSON Feeds

Phase Overview

Phase 1	Core Platform	Dashboard · Geo-Spatial · Graph · Timeline · Alerts · Data Explorer
Phase 2	Data Connectors	News Scraper · Beat Analytics · ANPR Tracker
Phase 3	AI Analysis	Entity Extraction · Relationship Discovery · Risk Scoring · Anomalies
Phase 4	Operations	Case Management · Live Tracking · Agency Fusion · Evidence Chain
Phase 5	Predictive	Hotspot Prediction · Disruption Simulation · Threat Forecast

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1. Phase 1 — Core Intelligence Platform

The foundation of ARGUS-NCR. A dark-theme military-grade intelligence dashboard with six interconnected modules built on Next.js 16 with TypeScript and Tailwind CSS.

1.1 Operations Dashboard

Real-time command center displaying the complete intelligence picture.

- 8 live stat cards: Total Entities, Active Alerts, Critical Alerts, Persons of Interest, Locations Monitored, Links Traced, Communications Intercepted, Financial Anomalies
- Risk distribution pie chart — Critical / High / Medium / Low breakdown with color-coded segments
- Entity distribution bar chart — categorization by type (Person, Location, Vehicle, Organization, Event, Communication, Financial)
- Live activity feed with timestamps and severity indicators
- Active alerts panel with acknowledgment tracking and severity filtering
- 30-second auto-polling for fresh data without page refresh

1.2 Geo-Spatial Analysis

Interactive dark-themed CartoDB map of the entire Delhi NCR region (13 police districts, 15+ key locations).

- Entity markers color-coded by type: Person (blue), Location (green), Vehicle (yellow), Organization (purple), Event (orange), Communication (cyan), Financial (red)
- Risk heat zones — pulsing overlay circles sized by risk score, color-coded red/orange/yellow
- Police district boundaries — 13 districts across Delhi, Gurugram, Noida, Ghaziabad, Faridabad
- Relationship lines — dashed lines connecting linked entities on the map
- Layer controls — filter by entity type, toggle heat zones, toggle district boundaries
- Click-to-inspect detail panel showing properties, tags, risk score, connected entities

1.3 Graph Analysis

Force-directed entity relationship graph with custom SVG physics simulation.

- 28 entities, 30 relationship links in the base network
- Node sizing varies by entity type, high-risk entities have red warning rings
- Hover to highlight connections and display link labels
- Click to open detail panel with full properties, risk gauge, tags, connections
- Type-based filtering — show/hide entity types to isolate networks
- Dark grid background with glow effects on interaction

1.4 Event Timeline

Chronological reconstruction of all timestamped intelligence events.

- Events grouped by date with severity-colored dots (Critical=Red, High=Orange, Medium=Yellow, Low=Green)
- Risk score display per event with visual gauge
- Property summaries and tag indicators inline
- Visual timeline rail with event markers for rapid scanning

1.5 Alerts Center

Dedicated alert management system with filtering and acknowledgment workflow.

- Severity-ordered alert cards (Critical !' Low) with color-coded borders
- Acknowledged/Unacknowledged sections for workflow tracking
- Timestamp, location coordinates, and linked entity references per alert
- Severity badges with iconography (Critical=Shield, High=AlertTriangle, etc.)
- 8 base alerts covering cross-border movement, hawala, encrypted comms, customs, land fraud, financial

1.6 Data Explorer

Full entity database with search, filter, sort, and expand capabilities.

- Search across name, type, tags, and property values
- Filter by entity type dropdown
- Sort by risk score (ascending/descending)
- Expandable rows showing properties and connections
- Inline risk score bar visualization
- Detail panel with full property dump, location, tags, connected entities

2. Phase 2 — Live Data Connectors

Three real-time data pipelines feeding live intelligence into the platform. Runs hourly via cron automation.

2.1 News Scraper

Real-time news aggregation from 5 Indian media RSS feeds focused on Delhi NCR.

- Sources: Times of India Delhi, The Hindu Delhi, Indian Express Delhi, NDTV Cities, Hindustan Times Delhi
- 15 real articles per scrape with title, description, publication date, and source
- Items displayed with source badges, timestamps, and clickable external links
- Auto-refresh every hour via cron job

2.2 Beat Analytics

Crime pattern simulation by police district across the entire NCR.

- 14 police districts covered: all 10 Delhi districts + Gurugram, Noida, Ghaziabad, Faridabad
- 15 crime types tracked: theft, burglary, assault, vehicle-theft, chain-snatching, cyber-fraud, domestic-violence, drug-possession, eve-teasing, snatching, robbery, riot, murder, kidnapping, extortion
- Per-capita incident rates calculated against district population
- Risk scores assigned based on incident count and crime type severity
- 7 reports generated per hourly cycle

2.3 ANPR Tracker

Simulated Automated Number Plate Recognition feed tracking vehicle movement.

- 8 tracked vehicles including known POI vehicles and unknown flagged plates
- 10 ANPR camera locations across Delhi NCR (toll plazas, junctions, border checkpoints)
- Vehicle details: plate number, make/model, color, registered owner
- Flagged vs un-flagged classification with confidence scores
- Direction tracking (Northbound/Southbound/Eastbound/Westbound)
- 5-10 hits per hourly cycle with 60-75% flag rate

3. Phase 3 — AI-Powered Analysis

Four AI engines processing Phase 2 data to extract intelligence, discover relationships, score risk dynamically, and detect anomalies.

3.1 Entity Extraction (NER)

Pattern-based Named Entity Recognition optimized for Indian names, Delhi NCR locations, and organization types.

- Person extraction: matches Indian first names + surnames against comprehensive name database (35+ first names, 35+ surnames)
- Organization extraction: pattern-matches company/organization keywords (Corp, Ltd, LLC, Group, Syndicate, Police, Bureau, Bank, etc.)
- Location extraction: matches against 25+ known Delhi NCR locations with geo-coordinates
- Processes all 15 Phase 2 news articles per cycle

3.2 Relationship Discovery

Co-occurrence analysis engine that finds hidden connections between entities.

- Cross-type co-occurrence: persons-to-organizations, persons-to-locations, orgs-to-locations
- Name-matching: detects when extracted entities match known database entities
- Confidence scoring: 55-95% based on co-occurrence frequency and pattern strength
- Evidence trail: links each discovered relationship back to source article

3.3 Dynamic Risk Scoring

Behavioral analysis model that recalculates entity risk scores based on real-time activity patterns.

- Factor 1 — Communication Volume: number of intercepted comms in 24h window
- Factor 2 — Movement Patterns: flagged ANPR hits per entity
- Factor 3 — Financial Activity: flagged transactions per entity
- Factor 4 — Network Position: HVT designation and known threat tags
- Factor 5 — News Exposure: mentions in scraped news articles
- Trend indicators: Escalating !‘ / Stable !’ / Declining !“
- 8 entities tracked with recalculated scores per cycle

3.4 Anomaly Detection

Statistical baseline deviation detection across four dimensions.

- Communication Spike: flags entities with 80%+ above-average comms volume
- Route Anomaly: flags vehicles detected at 3+ distinct locations (unusual movement)
- Financial Cluster: flags entities with 2+ flagged transactions in 24h
- Crime Cluster: flags districts with 4+ reported incidents (hotspot formation)
- Deviation percentage calculated against rolling baselines
- 3 anomalies detected per average cycle

4. Phase 4 — Operational Capabilities

Mission-ready operational tools for case management, live tracking, multi-agency coordination, and evidence chain custody.

4.1 Case Management

Active investigation tracking system with 5 open cases across multiple agencies.

- Case IDs with severity classification (Critical/High/Medium)
- Lead agency assignment with color-coded agency badges
- Entity linking — each case tracks associated persons, orgs, vehicles
- Status tracking with last update timestamps
- Case descriptions with operational context

4.2 Live GPS/ANPR Tracking

Real-time asset tracking with GPS coordinates, speed, heading, and status.

- 4 tracked assets: 2 vehicles + 2 field units
- Live GPS coordinates with refresh indicators
- Speed, heading, and last-ping timestamps
- Active/Idle status with pulsing green indicators
- Integration with ANPR camera network

4.3 Agency Fusion Center

Multi-agency coordination dashboard with 6 participating agencies.

- Agencies: Delhi Police (84,000 officers), NIA, Enforcement Directorate (2,000), Customs Intelligence (3,500), Intelligence Bureau (17,000), Cyber Cell (450)
- Per-agency case assignments with severity badges
- Secure comms indicator between agencies
- Total 359 active cases across all agencies

4.4 Evidence Chain

Custody chain tracking for all evidence items with multi-point verification.

- 5 evidence items tracked with unique IDs
- Evidence types: Phone Intercept, Document, Financial Record, Bank Statement, Property Deed
- Custody location tracking per evidence item
- Chain-of-custody counter (number of custodians who handled each item)
- Case linkage — each evidence item mapped to parent case

5. Phase 5 — Predictive Platform

Forward-looking analytical tools for hotspot prediction, network disruption simulation, and threat forecasting.

5.1 Hotspot Prediction

District-level risk forecasting with confidence scores and contributing factors.

- 6 districts under continuous prediction: Ghazipur Border Area (92/100), IGI Cargo (88/100), Chandni Chowk (85/100), Gurugram Cyber City (75/100), Dwarka (70/100), Jamia Nagar (68/100)
- Risk trend indicators: Rising/Stable with directional arrows
- Confidence scores: 68-85% based on data quality and pattern strength
- Contributing factors broken down per district (3-4 factors each)

5.2 Network Disruption Simulation

What-if scenario modeling — simulates the impact of arresting specific targets on the overall network.

- 3 simulation scenarios with varying target profiles
- Network fragmentation percentage — quantifies operation impact
- Downstream disruption listing — specific nodes affected
- Escaped node identification — operatives likely to evade
- Escalation risk assessment — Low/Medium/High with reasoning
- Coordinated raid scenario achieves 95% network fragmentation with high escalation risk

5.3 Threat Forecast

24/48/72-hour threat probability forecasts with actionable recommendations.

- Three timeframes: 24-hour (72% probability), 48-hour (64%), 72-hour (55%)
- Projected threats per timeframe (3 threats each)
- Recommended counter-actions per timeframe (3 actions each)
- Probability decreases over time reflecting increasing uncertainty
- Action-items formatted as tactical directives for operational teams

6. Dataset & Intelligence Scenario

The ARGUS-NCR platform runs on a realistic, internally consistent intelligence scenario centered on organized crime networks in Delhi NCR.

6.1 Entity Network

28 base entities distributed across 7 types:

- 8 Persons of Interest: Vikram Malhotra (Risk 78), Rashid Qureshi (Risk 91 — HVT), Anita Sharma (Risk 45), Deepak Tyagi (Risk 22), Farhan Siddiqui (Risk 85), Meera Kapoor (Risk 15), Suresh Kalmadi (Risk 67), Priya Nair (Risk 52)
- 4 Locations: Ghaziabad Warehouse (Risk 72), Jamia Nagar Safe House (Risk 88), Chandni Chowk Import Office (Risk 65), IGI Airport Cargo Terminal (Risk 40)
- 3 Vehicles: DL-3C-AX-7842 (Innova, Malhotra), UP-16-T-9054 (Tata Ace, Qureshi), HR-26-AB-1234 (Creta, Kalmadi)
- 3 Organizations: Siddiqui Imports LLC (Risk 82), NCR Transport Syndicate (Risk 75), Delhi Infrastructure Corp (Risk 58)
- 4 Events: Cargo Shipment, Cash Deposit, Border Crossing, Property Registration
- 3 Communications: Phone Intercept, Email Intercept, WhatsApp Group
- 2 Financial: Hawala Transfer, Property Payment

6.2 Key Scenario Threads

Four interconnected criminal networks form the intelligence picture:

- Smuggling Pipeline: Rashid Qureshi !’ cross-border movement !’ Ghaziabad ware Imports — 30 relationship links connecting the network
- Hawala Network: ¹3.2Cr traced from Chandni Chowk to Dubai, linked to Malhotr suspected
- Corruption Ring: Anita Sharma (MCD official) + Suresh Kalmadi (developer) !’ s Delhi Infrastructure Corp — property registration in Dwarka flagged
- Encrypted Comms: "Project Alpha" WhatsApp group with 7 members and 234 messages coordinating smuggling operations — decrypted by signals intelligence

6.3 Network Connectivity

30 relationship links connecting all entities into a cohesive intelligence graph:

- Communication links: Malhotra !” Qureshi (high frequency), Qureshi !” Siddiqui (
- Business links: Malhotra + Siddiqui are directors of Siddiqui Imports LLC
- Employment links: Anita Sharma + Suresh Kalmadi linked to Delhi Infrastructure Corp
- Ownership links: Vehicles traced to Malhotra, Qureshi, Kalmadi
- Event links: Cargo shipment tied to Siddiqui Imports, border crossing tied to Qureshi vehicle, property registration involving Sharma and Kalmadi

7. Automation & Scheduling

All data pipelines run on cron-based schedules ensuring the platform stays current without manual intervention.

7.1 Cron Jobs

- Daily Feed Generator: Runs at 6:00 AM IST daily — generates 10-15 new intelligence signals (events, comms, financial, alerts)
- Phase 2 Connectors: Runs every hour — scrapes 15 news articles, generates 7 beat reports, tracks 5-10 ANPR hits
- Phase 3 AI Engine: Runs every hour — processes Phase 2 output through entity extraction, relationship discovery, risk scoring, and anomaly detection

7.2 Data Flow Pipeline

Phase 2 Connectors !' phase2-feed.json (public/) !' Phase 3 Engine reads phase2-fee

Dashboard polls daily-feed.json every 30 seconds

Phase 2 view polls phase2-feed.json every 60 seconds

Phase 3 view polls phase3-feed.json every 60 seconds

No rebuild required — all feeds served as static JSON from the public/ directory

7.3 Output Files

- daily-feed.json — Synthetic daily intelligence (movement, comms, financial, alerts)
- phase2-feed.json — Live scraped news, beat reports, ANPR hits
- phase3-feed.json — Entity extraction results, discovered relationships, recalculated risk scores, anomaly flags

8. Technical Architecture

8.1 Frontend Stack

- Framework: Next.js 16 with App Router and TypeScript
- Styling: Tailwind CSS with custom CSS variables for dark military-grade theme
- Maps: Leaflet with CartoDB dark tiles, dynamic import for SSR safety
- Charts: Recharts (PieChart, BarChart with dark theme customization)
- Graph: Custom SVG force-directed layout with velocity-based physics simulation
- Icons: Lucide React (60+ icons used across platform)
- PDF: jspdf for documentation generation

8.2 Backend & Data

- Data Pipelines: Node.js scripts (ES modules) with crypto for ID generation
- Web Scraping: Native http/https modules + regex-based RSS parsing
- NLP: Pattern-based NER with Indian name databases (70 names) and Delhi NCR gazetteer
- Analytics: Statistical deviation detection with rolling baselines and multiplicative thresholds
- Scheduling: OpenClaw cron system with isolated agent sessions

8.3 File Structure

Key files in the ARGUS-NCR codebase:

- src/lib/data.ts — Base entity network (28 entities, 30 links, 8 alerts)
- src/lib/daily-feed.ts — Daily intelligence feed (auto-generated)
- scripts/generate-feed.js — Daily feed generator
- scripts/phase2-connectors.js — News scraper + beat analytics + ANPR tracker

- scripts/phase3-engine.js — Entity extraction + relationship discovery + risk scoring + anomaly detection
- public/daily-feed.json — Runtime feed (no rebuild required)
- public/phase2-feed.json — Phase 2 runtime feed
- public/phase3-feed.json — Phase 3 runtime feed
- src/components/ — 12 React components (sidebar, dashboard, geospatial, graphview, timeline, alertsview, dataview, phase2-view, phase3-view, phase45-view, ouroboros-logo)

8.4 Performance

- Static exports: JSON feeds served directly from public/ directory — zero server processing
- Client-side polling: 30-60 second intervals with cache-busting params
- Map lazy loading: Leaflet dynamically imported in useEffect to avoid SSR window errors
- Graph physics: Client-side force simulation with 200 iterations, capped at 40 nodes
- Build size: ~37KB initial page, feeds under 50KB each

ARGUS-NCR

Delhi NCR Intelligence Platform

Platform Statistics

- 28 Base Entities · 30 Relationship Links · 8 Base Alerts
- 5 Phase Pipeline: Core !' Data !' AI !' Ops !' Predictive
- 12 React Components · 3 Node.js Engine Scripts
- 3 Cron Jobs · 3 Live JSON Feeds · 7 Data Modules
- 15 Real-time News Articles (Hourly) · 5 RSS Sources
- 14 Police Districts · 15+ Key Locations · 15 Crime Types
- 10 ANPR Camera Locations · 8 Tracked Vehicles
- 6 Participating Agencies · 5 Active Cases · 5 Evidence Items
- 6 Predictive Hotspots · 3 Disruption Scenarios · 3 Threat Forecasts

Classification: UNCLASSIFIED | Demonstration Model with Synthetic Data
Stack: Next.js 16 + TypeScript + Tailwind + Leaflet
No real intelligence, PII, or classified information is included.